
GIT PUSH THE BEATS

Melody Dive **Use-Case Specification: Manage music catalog**

Version 1.0

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Use-Case Specification: Manage music catalog	Date: 15/11/2025
UC Manage music catalog	

Revision History

Date	Version	Description	Author
15/11/2025	1.0	The initial version	Huỳnh Gia Bảo

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1. Use-Case Name: Manage music catalog

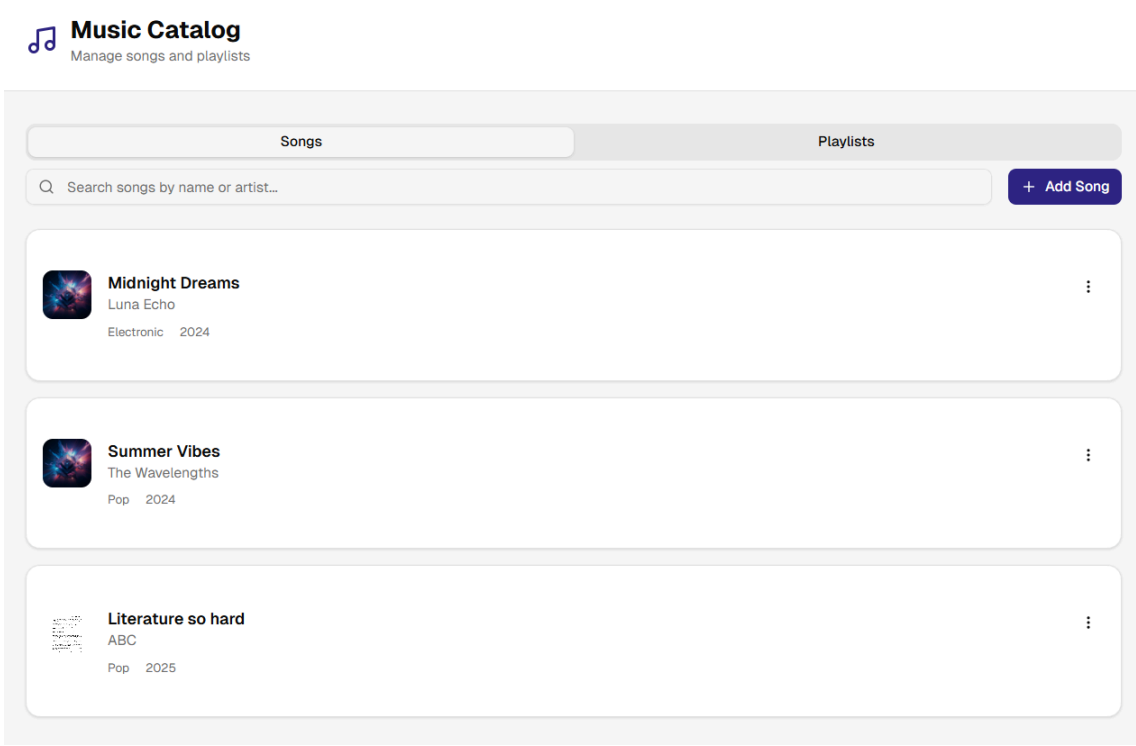
1.1 Brief Description

This use-case describes how admins can add, remove, and modify songs and playlists of songs.

2. Flow of Events

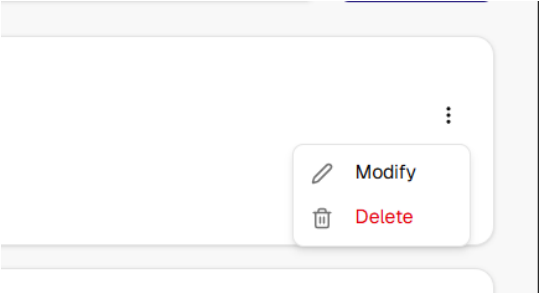
2.1 Basic Flow

- 1. Admin selects *Manage music catalog* section.
- 2. Admin can see 2 sections: *Songs* and *Playlists*, choose one section to view all songs/playlists.



3. Manage songs

- Remove a song:
 - + Admin selects a song to view options
 - + Then, admin chooses **Delete song**.
 - + Then, the system deletes the song from the platform.



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- Add a song:
 - + Admin selects **Add song** to add song's audio file, its avatar picture and its metadata (name, author's name, composer's name, genre, published year, lyrics, etc).
 - + If the song already exists, it goes to the *Existed song* alternative flow mentioned below, or else the system will receive the data and create the song according to those data.

Add Song

Song Avatar
Choose File No file chosen

Song Name
Enter song name

Artist
Enter artist name

Composer
Enter composer name

Genre **Year**
e.g., Pop, Rock 2025

Music File (MP3, WAV, etc.)
Choose File No file chosen

Plain Lyrics
Enter song lyrics

Synced Lyrics (with Timestamps)
Upload Lyrics File (LRC or TXT)

- Modify a song:
 - + Admin selects a song to view options.
 - + Admin then chooses **Modify song** to modify its metadata, or to change its avatar picture.
 - + The system updates the song.

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Modify Song

Song Avatar

Choose File No file chosen

Song Name

Midnight Dreams

Artist

Luna Echo

Composer

Alex Moon

Genre

Electronic

Year

2024

Music File (MP3, WAV, etc.)

Choose File No file chosen

Plain Lyrics

Sample lyrics...

Synced Lyrics (with Timestamps)

Upload Lyrics File (LRC or TXT)

4. Manage playlists

- Remove a playlist:
 - + Admin selects a playlist to view options.
 - + Admin then chooses **Delete playlist**.
 - + The system deletes the playlist.

Songs

Playlists

ronic music

Summer Hits

Best hits of the season

2 songs

Modify

Delete

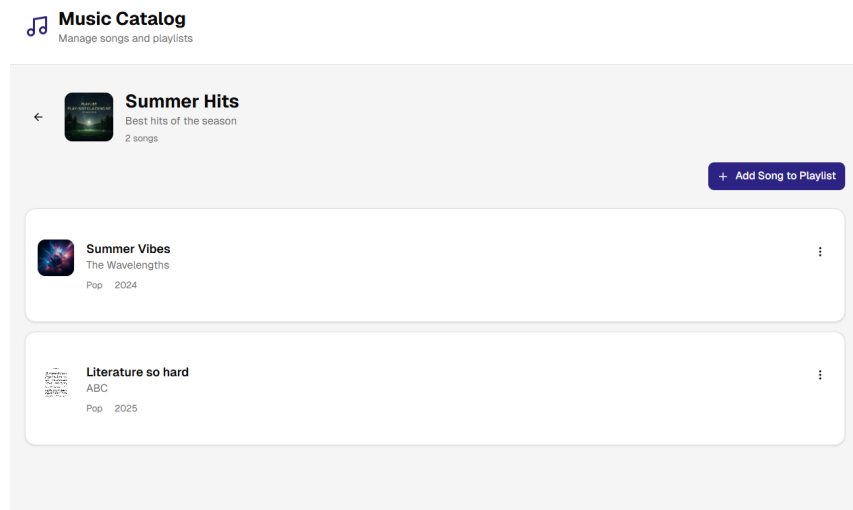
- Add a playlist:
 - + Admin selects **Add playlist** to create a new playlist.
 - + Admin then names it and uploads its avatar picture.
 - + The system checks if the playlist’s name already exists. If not, the playlist is generated.

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- Modify a playlist:
 - + Admin selects a playlist to view options.
 - + Admin then chooses **Modify playlist** to modify it. Specifically, admin can change its name, and avatar picture.
 - + The system checks if the playlist's name already exists. If not, the playlist is updated.

- Manage songs in a playlist:
 - + Admin goes to a specific playlist to view all its songs
 - + Then, admin follows the steps mentioned above in **Manage song**, the flow will be the same as managing song.

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2.2 Alternative Flows

2.2.1 Existed song/playlist

This alternative flow starts when the admin adds a song/playlist which already existed, or when the admin modifies a song/playlist then makes it become an existed one.

A song is defined ‘existed’ if its metadata already exists., and a playlist is defined ‘existed’ if its name already exists.

In this flow, an error message will be displayed and the song/playlist will not be created/modified.

3. Special Requirements

3.1 Speed

The time to upload a song must be less than a quarter of the song’s duration. (eg: Uploading a song lasting 4 minutes must take less than 1 minute.)

Metadata takes less than 1 second to be updated.

3.2 Concurrence

The system allows up to 10 admins to upload songs at the same time, and still ensures the speed mentioned above.

4. Preconditions

4.1 Admin account logged in

The admin must have logged in an **Admin** account.

5. Postconditions

5.1 Song/playlist updated

The song/playlist is added/removed/modified.